

University of Reading School of Pharmacy



PM3RP: Research Project

**Docking Reproducibility: *Vina*, *Smina* and *GOLD*
Compared**

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This project is submitted in partial fulfilment of the requirements for the BSc Pharmacology. It describes work carried out in the Department of Pharmacy during the academic year 2025-26. Unless otherwise indicated, the research described is my own and not the product of collaboration.

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ABSTRACT

Structure-based drug design relies on molecular docking to approximate protein–ligand binding conformations and relative interaction stability. Although engines such as *AutoDock Vina* and *Smina* are widely used in academic workflows, systematic evaluation of their reproducibility under regulated parameter alterations remains limited. In practice, docking reliability is often assumed rather than critically examined. This study investigates docking reproducibility as an emergent property of engine–target interaction. Three structurally distinct crystallographic systems, HIV-1 protease (1HVR), COX-2 (4PH9), and β -trypsin (3PTB), were used to evaluate rigid and flexible redocking, parameter sensitivity to exhaustiveness, grid size, and stochastic seed variation, and ligand-preparation perturbations. Commercial validation had been performed using *GOLD* (ChemPLP) on the 3PTB system.

Rigid redocking consistently produced lower RMSD values than flexible protocols over all three systems, establishing it as a stable convergence baseline. Parameter sensitivity differed markedly between systems: *Vina* showed strong convergence stability for 1HVR, whereas both engines exhibited pronounced variability for 3PTB under altered grid and exhaustiveness conditions. *GOLD* attained comparable pose accuracy to both open-source engines under matched conditions. Ligand-preparation effects were also target-dependent. This evidence shows that docking reproducibility is not an intrinsic property of a software engine but depends on the structural and energetic topography of the target protein. Reliable use of open-source docking, therefore, requires target-specific validation rather than reliance on default settings. This has clear effects on virtual screening workflows and on the Quality of training data used in machine-learning scoring functions, where target-dependent irregularities in the underlying docking data risk being carried forward into the model.

List of Abbreviations

Å	ÅNGSTRÖM
1HVR	HIV-1 Protease Protein Data Bank identifier
3PTB	β-trypsin–benzamidine Protein Data Bank identifier
4PH9	COX-2 Protein Data Bank identifier
ADT	AutoDockTools
ANOVA	Analysis of Variance
CCDC	Cambridge Crystallographic Data Centre
CHEMPLP	Piecewise Linear Potential (<i>GOLD</i> scoring function)
COX-2	Cyclooxygenase-2
CPU	Central Processing Unit
CV	Coefficient of Variation
GA	Genetic Algorithm
GNINA	Machine learning-augmented molecular docking software
GOLD	Genetic Optimisation for Ligand Docking
GPU	Graphics Processing Unit
H-BOND	Hydrogen bond
HIV-1	Human Immunodeficiency Virus type 1
IBP	Ibuprofen
KCAL/MOL	Kilocalories per mole
MMFF94	Merck Molecular Force Field 1994
NMR	Nuclear Magnetic Resonance
PDB	Protein Data Bank
PDBQT	Protein Data Bank format with partial charges (Q) and atom types (T)
PYMOL	Molecular visualisation software
RMSD	Root Mean Square Deviation
SMINA	Modified <i>AutoDock</i> Vina docking engine
VINA	<i>AutoDock</i> Vina molecular docking engine
WSL	Windows Subsystem for Linux
XK2	Cyclic urea inhibitor (ligand in 1HVR)

1. INTRODUCTION

Drug molecules exert their therapeutic effects by binding to specific protein targets in the body. The strength of this interaction, its selectivity for the intended target, and the resulting three-dimensional binding mode collectively determine efficacy and safety (Hughes *et al.*, 2011). Small-molecule compounds remain the foundation of modern pharmacotherapy and account for the majority of approved drugs. Their recognition reflects favourable physicochemical properties, often summarised by Lipinski's Rule of Five: molecular weight ≤ 500 Da, appropriate lipophilicity, and suitable hydrogen-bonding capacity, which together support oral bioavailability and systemic distribution. In contrast, larger biologics frequently require parenteral administration. Accurate and reproducible prediction of binding mode and affinity is therefore critical in early drug discovery. If docking results vary due to parameter choices rather than genuine binding differences, compounds may be incorrectly prioritised or discarded, wasting resources and potentially excluding viable therapeutic candidates (Kitchen *et al.*, 2004). This concern forms the central motivation for the present study.

1.1 Characterising Protein-Ligand Complexes

When a ligand binds to its protein target, a protein–ligand complex is formed. Determining the atomic structure of such complexes provides essential insight into molecular recognition and the mechanism of drug action. X-ray crystallography remains the primary experimental method for determining fine structure. In this technique, purified proteins are crystallised and exposed to intense X-ray beams, typically generated at synchrotron facilities. The resulting diffraction patterns arise from constructive and destructive interference of X-rays scattered by electrons inside the crystal lattice. Mathematical reconstruction of these patterns by solving the crystallographic phase problem yields an electron density map from which atomic coordinates are modelled (Garman & Owen, 2007). All three systems examined in the present study were derived from X-ray crystallographic data deposited in the PDB.

1.2 The Protein Data Bank

The Protein Data Bank (PDB) is the primary global repository for experimentally determined biomolecular structures and currently contains over 220,000 entries, including many pharmacologically relevant protein–ligand complexes. As an open-access resource, the PDB provides atomic coordinates, experimental data such as electron density maps, and metadata including resolution and refinement statistics in standardised formats appropriate for computational analysis (Berman, 2000). The PDB underpins structure-based drug design by supplying experimentally validated templates for computational modelling, docking, and molecular simulation studies (Sadybekov & Katritch, 2023). All three protein systems evaluated in the present study were selected from PDB entries based on resolution and the availability of co-crystallised ligands.

1.3 Computational Approaches to Docking

Molecular docking uses computational methods to predict how small molecules bind within a defined protein binding site. It is used both when experimental ligand-bound structures are unavailable and in virtual screening workflows, where large chemical libraries can be evaluated efficiently (Trott & Olson, 2010). Docking has two main components. The first is a search algorithm, which samples ligand conformations, orientations, and translations within a defined binding region. The second is a scoring function, which estimates binding affinity using a simplified energy model and ranks the resulting poses. Although docking is widely used, it remains an approximation. Real protein–ligand binding occurs on a multidimensional energy landscape formed by steric complementarity, electrostatics, solvation, and conformational entropy. Docking engines simplify this landscape through empirical scoring functions and stochastic sampling approaches. As a result, docking reliability depends not only on the software itself but also on how effectively the search strategy locates the best pose on the scoring surface.

The topology of the scoring surface differs between targets. Proteins with deep, highly constrained pockets may present smooth energy basins, enabling rapid convergence. In contrast, binding sites with multiple subpockets, conformational flexibility, or competing interaction modes may produce rough landscapes containing numerous local troughs. In such cases, stochastic selection parameters such as exhaustiveness and random seed can strongly influence convergence behaviour. Improvements in search efficiency, scoring function design, and computational hardware have enabled virtual high-throughput screening (Lionta *et al.*, 2014). However, predicted binding energies correlate only moderately with actual affinities, and outcomes depend strongly on preparation and parameter choice (Kitchen *et al.*, 2004). Understanding docking as an interaction between sampling depth and landscape topology is therefore essential when evaluating reproducibility.

1.4 Docking Strategies and Codes

Although all molecular docking programs seek to predict ligand binding modes within a protein's active site, they vary fundamentally in how they estimate binding energy and explore conformational space. Docking performance emerges from the interaction between these two components rather than from either one in isolation.

1.5 Scoring Functions and Force Fields

Docking scoring functions are simplified mathematical models designed to approximate changes in binding free energy upon ligand association. True binding free energy reflects enthalpic and entropic contributions, solvent reorganisation, conformational strain, and long-range electrostatics. Accurate calculation of these components using rigorous physical methods is computationally too expensive for routine screening. Most docking engines therefore use empirical scoring functions that approximate binding through weighted interaction terms, typically including van der Waals

complementarity, hydrogen bonding, electrostatic interactions, hydrophobic contacts, and torsional strain penalties. The weighting of these terms is usually derived from regression against experimentally characterised protein–ligand complexes. As a result, docking scores should be interpreted as relative ranking metrics rather than absolute thermodynamic values. The three principal categories of scoring functions, and the placement of the tools used in the present study within this taxonomy, are summarised in Figure 1.

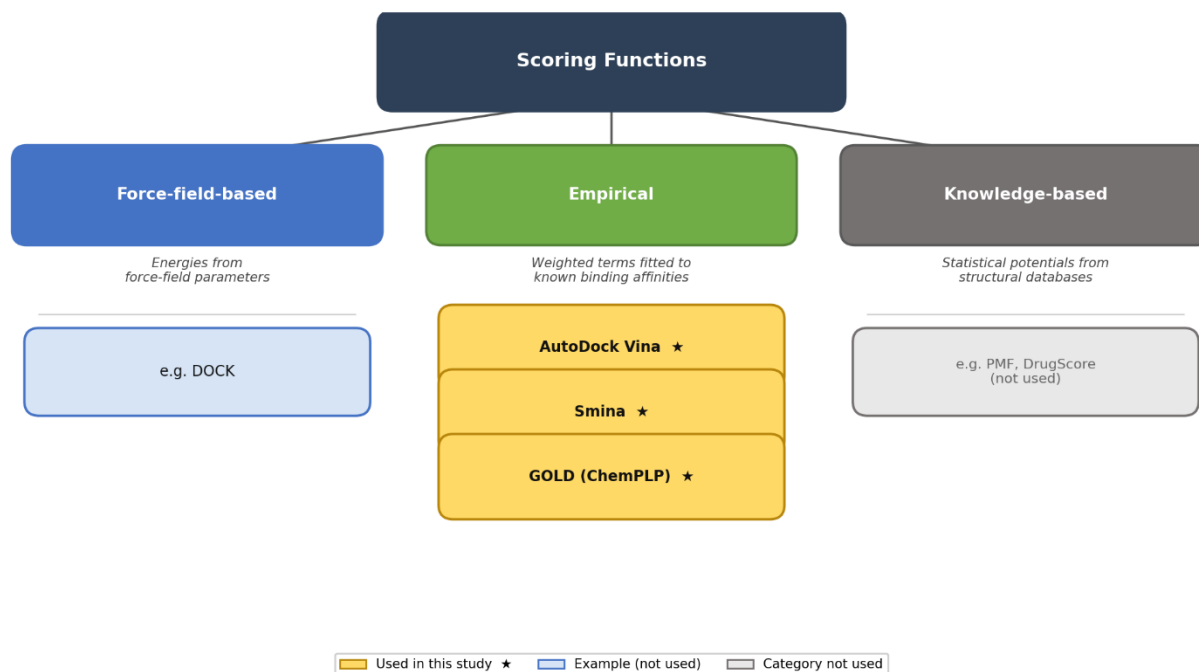


Figure 1. Classification of molecular docking scoring functions into force-field-based, empirical, and knowledge-based categories. *AutoDock Vina*, *Smina* and *GOLD* (ChemPLP) all employ empirical scoring functions in which interaction terms are derived by regression against experimentally characterised protein–ligand complexes.

AutoDock Vina employs a compact empirical scoring function designed for speed and reliable pose prediction (Trott & Olson, 2010). *Smina* retains the same framework but allows scoring parameters to be modified (Koes, Baumgartner and Camacho, 2013) and supports alternative scoring schemes such as Vinardo. While this flexibility is useful for benchmarking and method development, it can also increase sensitivity to parameter configuration. Because scoring functions are approximations, docking reliability depends not only on their design but also on whether the correct pose is sampled during the search process.

1.6 Search Algorithms

These search algorithms include user-controllable parameters that can greatly affect docking performance and reproducibility. Docking explores a high-dimensional conformational space defined by ligand translations, rotations, and torsional degrees of freedom, so exhaustive enumeration is infeasible and stochastic search strategies are required. *AutoDock Vina* and *Smina* employ hybrid search methods that combine global random selection within a user-defined grid box with local gradient-based optimisation toward nearby energy minima. The exhaustiveness parameter governs the number of independent search attempts and therefore the sampling depth. values increase the probability of locating the global minimum, but at a greater computational cost. Figure 2 illustrates the wider classification of flexible docking search algorithms, situating *AutoDock Vina*, *Smina*, and *GOLD* within their respective methodological branches, alongside *FlexX* as a contrasting example of incremental construction.

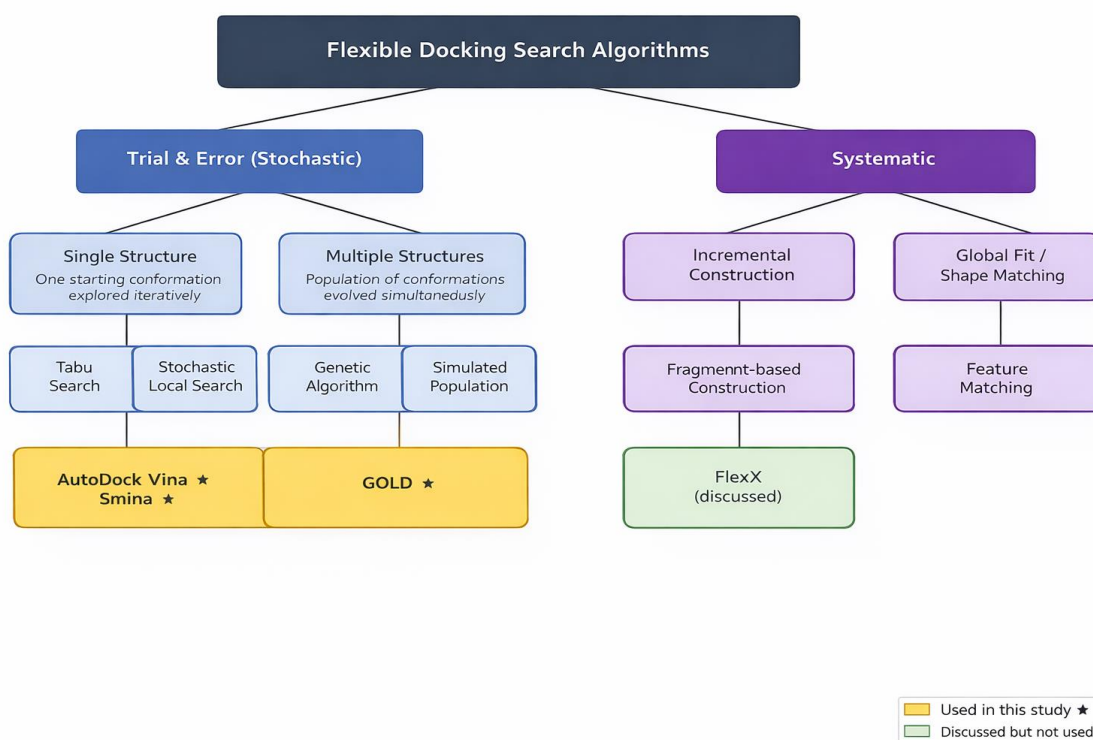


Figure 2. Classification of flexible docking search algorithms. AutoDock Vina and Smina use stochastic local search (single structure), GOLD uses a genetic algorithm (multiple structures), and FlexX uses fragment-based incremental construction (systematic branch) (Kellenberger et al., 2004).

The grid box defines the spatial boundaries of exploration. Overly restrictive grids may exclude relevant poses, whereas excessively large grids increase processing burden and may reduce search efficiency. The random seed determines the initial sampling configuration, allowing reproducible or intentionally varied runs. Differences between seeds may reveal sensitivity to local minima or incomplete sampling. Docking accuracy, therefore, depends on how effectively the search strategy explores the scoring landscape: a scoring function cannot recognise a correct pose

if it is never sampled, and extensive sampling is of limited value if the scoring function cannot distinguish near-native geometries.

Despite algorithmic improvements, no stochastic search guarantees identification of the global minimum within a finite computational time. The balance between sampling depth and processing cost, therefore, becomes a practical constraint in routine workflows. Inadequate exhaustiveness may lead to premature convergence on local minima, whereas excessive sampling increases runtime without necessarily improving ranking reliability. Consequently, search parameter choice represents a trade-off between convergence stability and computational effectiveness, and it may influence reproducibility across structurally diverse targets.

1.7 Available Codes

Docking engines differ in scoring philosophy, search strategy, and intended use context. *AutoDock Vina* and *Smina* both use stochastic local search, beginning from a single conformation and refining iteratively through local optimisation (Trott & Olson, 2010; Koes, Baumgartner and Camacho, 2013). Because each run starts from one initial point, outcomes can vary depending on the random seed, a sensitivity directly examined in the present study. *GOLD*, by contrast, uses a genetic algorithm, maintaining a population of poses that evolve concurrently through mutation and crossover, delivering broader conformational coverage and potentially greater robustness against local minima (Jones et al., 1997). *FlexX* takes a different approach, using incremental construction to place a segment anchor before sequentially building the ligand within the pocket, although early placement errors can propagate through the structure. A key consequence of these architectural differences is that convergence behaviour is not determined by software choice alone, but by how each search strategy interacts with the energy surface of a given binding site. Extensive evaluation, therefore, requires systematic testing under controlled testing conditions rather than reliance on default settings or reputation.

1.8 Methodological Challenges in Docking Workflows

Given the large number of docking engines, scoring functions, and search algorithms currently available, each with different input requirements and default settings, there is a clear need for guidance on the practical strengths and limitations of these approaches. In most academic workflows, the commonly used tools are not complete graphical packages but a combination of command-line programs and separate visualisation software. *AutoDockTools* is typically used to prepare receptors, ligands (Morris *et al.*, 2009), generate PDBQT files, define grid boxes, and inspect docked poses, while *PyMOL* and *Chimaera* are used mainly for structure visualisation, with *Chimaera* also offering basic interaction analysis despite its older and less intuitive interface. Commercial suites package many of these steps into integrated graphical interfaces, but licensing costs can limit their use. By contrast, open-source engines such as *AutoDock Vina* and *Smina* are

freely available but rely heavily on scripting, manual file handling, and detailed parameter setting, which can create a steep learning curve for inexperienced users.

Despite docking's proven utility, academic researchers still face several practical challenges when using open-source engines such as *AutoDock Vina* and *Smina*. (Trott and Olson, 2010; Koes et al., 2013). These tools rely heavily on rigorous parameter configuration, yet there is no universal agreement on the best settings for grid box placement, exhaustiveness, receptor flexibility, or pose sampling among different protein–ligand systems. Ligand preparation steps, such as energy minimisation and protonation state assignment, can also cause variability that affects both docking scores and pose reliability. Furthermore, most published studies focus on benchmarking absolute performance rather than systematically exploring how small, controlled parameter changes modify outcomes in routine workflows. The present study, therefore, treats reproducibility as a measurable, target-dependent property rather than a binary assumption, and investigates it by controlled variation of search depth, grid definition and ligand preparation across three structurally diverse systems.

2. AIMS AND OBJECTIVES

Aim

To evaluate and optimise molecular docking workflows for predicting small-molecule ligand–receptor interactions using widely used open-source docking engines.

Objectives

1. To examine how key docking parameters (grid box placement and size, exhaustiveness, and random seed) influence predicted binding energies in *AutoDock Vina* and *Smina*.
2. To assess the impact of simple ligand modifications on docking scores and pose reproduction under otherwise identical docking conditions.
3. To compare *AutoDock Vina* and *Smina* using experimentally characterised protein–ligand complexes, focusing on their ability to reproduce crystallographic binding poses and their relative scoring behaviour within each engine.

It was hypothesised that docking reproducibility would vary between protein targets and increase in sensitivity under flexible ligand treatment and altered search parameters.

3. MATERIALS AND METHODS

3.1 Software

The following software tools were selected to provide a modular workflow combining open-source and commercial solutions, each chosen for its specific role in the docking pipeline.

Software	Version	Purpose
<i>AutoDock Vina</i>	1.2.7	Primary docking engine
<i>Smina</i>	2.0	Alternative docking engine
<i>GOLD (ChemPLP)</i>	v2026	Commercial validation
<i>AutoDockTools (ADT)</i>	1.5	Receptor/ligand preparation; PDBQT generation
<i>UCSF ChimeraX</i>	1.10	Structural inspection; interaction analysis
<i>PyMOL</i>	3.1.x	Structure superposition; pose visualisation

ADT was used for receptor and ligand preparation. Docking runs were executed via command line using *Vina* and *Smina* configuration files. PyMOL was used for posing, alignment, and RMSD calculation. ChimeraX was used for hydrogen bond and contact analysis.

3.2 Hardware

All docking calculations were performed on a standard desktop computer.

Component	Specification
CPU	AMD Ryzen 5 5500U (6-core/12-thread, 2.1–4.0 GHz)
RAM	16 GB DDR4
Operating System	Windows 11
Storage	512 GB SSD

Multi-core CPU exploited built-in multithreading of *AutoDock Vina*, *Smina*, and *GOLD*, enabling parallel pose optimisation across docking experiments.

3.3 Test Systems

Three crystallographic protein–ligand complexes were selected from the Protein Data Bank as benchmark systems.

Cyclooxygenase-2 (COX-2) is a membrane-associated enzyme that catalyses the rate-limiting step in prostaglandin biosynthesis and represents a major therapeutic target for anti-inflammatory drug development (Vane, Bakhle and Botting, 1998). The COX-2/ibuprofen complex 4PH9 was selected as the third benchmark system on the basis of its binding site character: the pocket is well-defined but moderately sized, with ibuprofen accommodated through a combination of hydrophobic contacts and a small number of hydrogen-bonding interactions (Kurumbail *et al.*, 1996; Orlando, Lucido and Malkowski, 2015). This places COX-2 in a structurally intermediate position relative to the other two targets, being less electrostatically constrained than the S1 pocket of 3PTB and less expansively hydrophobic than the cavity of 1HVR, which made it useful for assessing whether target-dependent reproducibility follows a graded pattern across binding-site architectures (Kitchen *et al.*, 2004). COX-2 exists as a homodimer; docking was performed on chain A only, with the crystallographic ibuprofen pose serving as the reference.

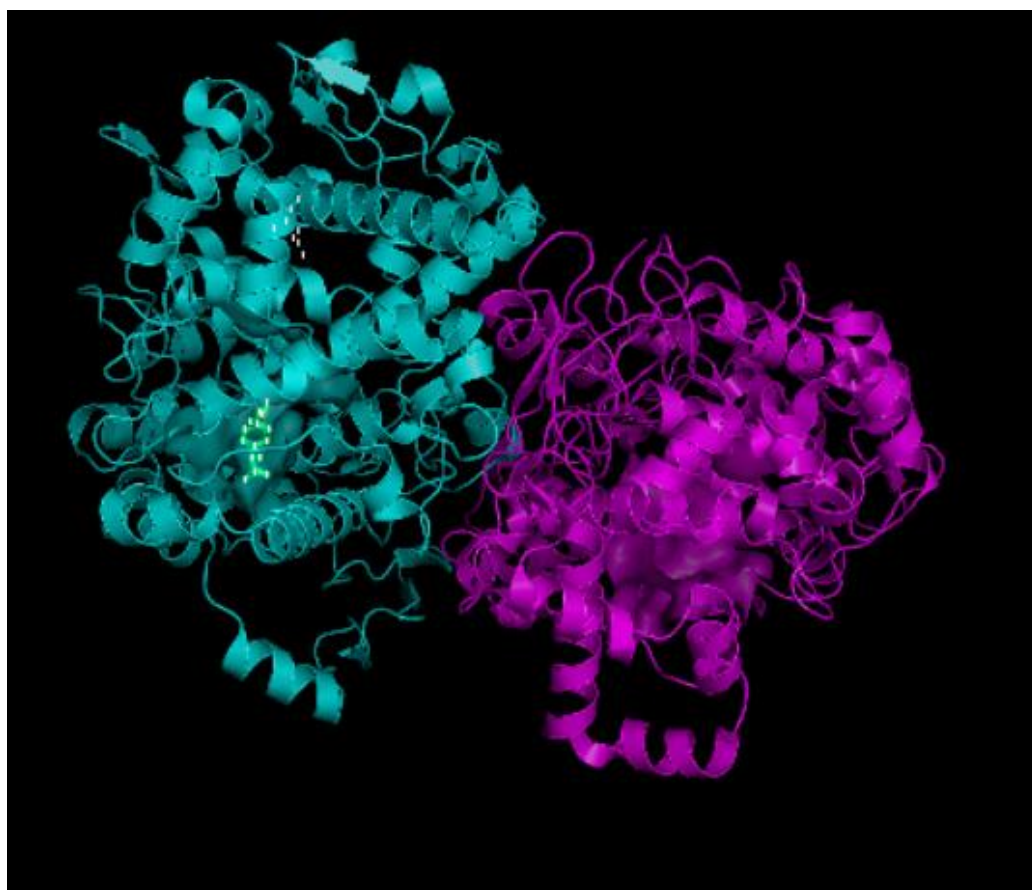


Figure 3. Cartoon representation created in *PyMOL* of the COX-2 homodimer (PDB ID: 4PH9). Chain A is shown in cyan and chain B in magenta, with the co-crystallised ibuprofen ligand (IBP) displayed as green sticks within the active-site cavity of chain A. Docking was performed on chain A only.

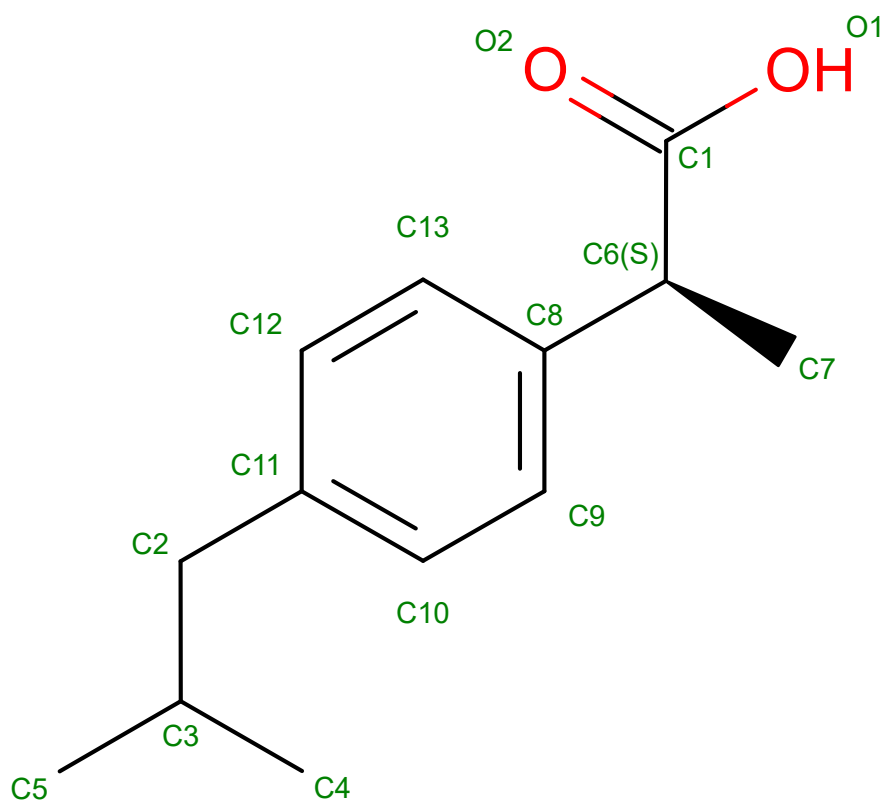


Figure 4. Two-dimensional chemical structure of ibuprofen (COX-2 ligand, IBP). The carboxylic acid group mediates hydrogen bonding within the active site, while the isobutyl and methyl substituents contribute hydrophobic contacts within the binding pocket.

HIV-1 protease is a clinically validated antiviral target whose inhibitors form a cornerstone of antiretroviral therapy (Eron, 2000). Unlike the electrostatically constrained S1 pocket of trypsin, its active site is a broad hydrophobic cavity that accommodates distributed ligand contacts across multiple residues, a fundamentally different binding environment, and one that comes with its own complications. The flap regions flanking the active site are conformationally flexible, which brings dynamic variability that simpler rigid-pocket systems do not capture. Numerous high-resolution inhibitor complexes exist in the PDB, including 1HVR, making it a well-established benchmark for docking and scoring evaluation (Lam *et al.*, 1994; Kaldor *et al.*, 1997; Kitchen *et al.*, 2004). It was chosen here as a direct contrast to 3PTB, where benzamidine docking is governed by a single dominant electrostatic interaction. HIV-1 protease was expected to produce more stable pose convergence across parameter conditions, allowing any differences in reproducibility to be attributed to target character rather than noise.

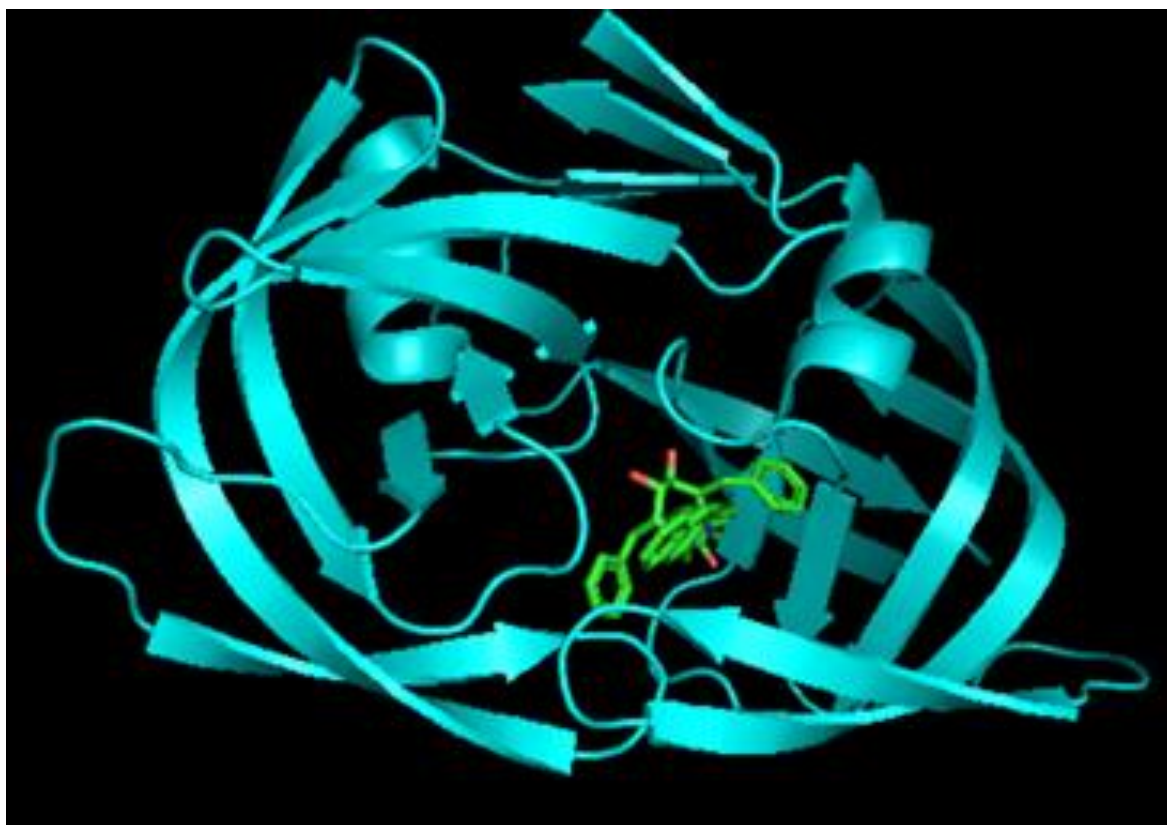


Figure 5. HIV-1 protease (PDB ID: 1HVR) structure. Cartoon representation created in *PyMOL* of the dimeric HIV-1 protease shown as a cyan ribbon, with the co-crystallised cyclic urea inhibitor (XK2) displayed as green sticks within the central active-site cavity.

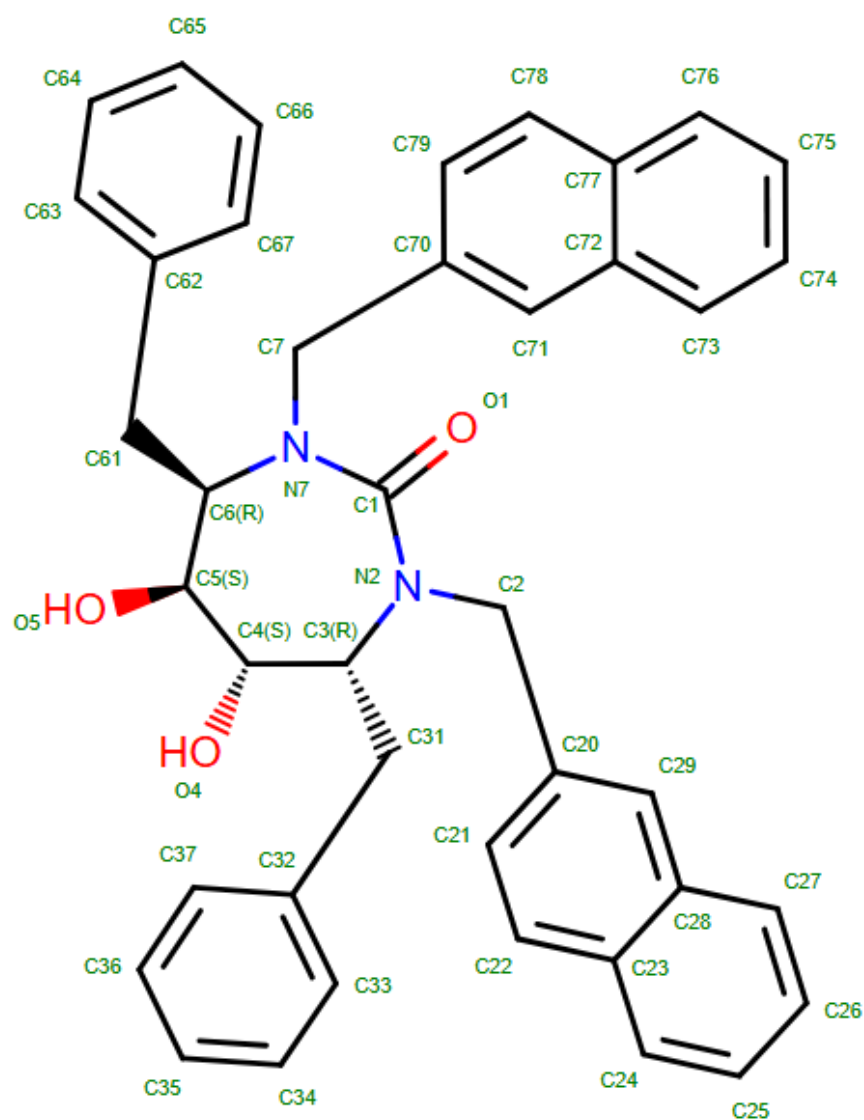


Figure 6. Two-dimensional chemical structure of XK2 cyclic urea inhibitor (1HVR ligand). The extended hydrophobic arms and multiple rotatable bonds reflect the structural complexity of the molecule relative to benzamidine.

β -Trypsin is a serine protease with a well-defined S1 specificity pocket and a large number of crystallographic inhibitor complexes in the PDB, including the widely cited benzamidine–trypsin structure 3PTB. It is not a primary therapeutic target in contemporary drug discovery, but it has become a standard methodological benchmark precisely because of its structurally constrained binding site and relatively simple ligand systems such as benzamidine (Kitchen *et al.*, 2004; Huang, 2018). The S1 pocket is electrostatically dominated by Asp189 at its base, which forms the key interaction with the positively charged benzamidine group, meaning successful pose reproduction hinges almost entirely on correct ligand orientation relative to a single residue. That constraint is what makes it a useful test case here: the system is simple enough to isolate the effect of parameter choices without ambiguity about what a correct result looks like.

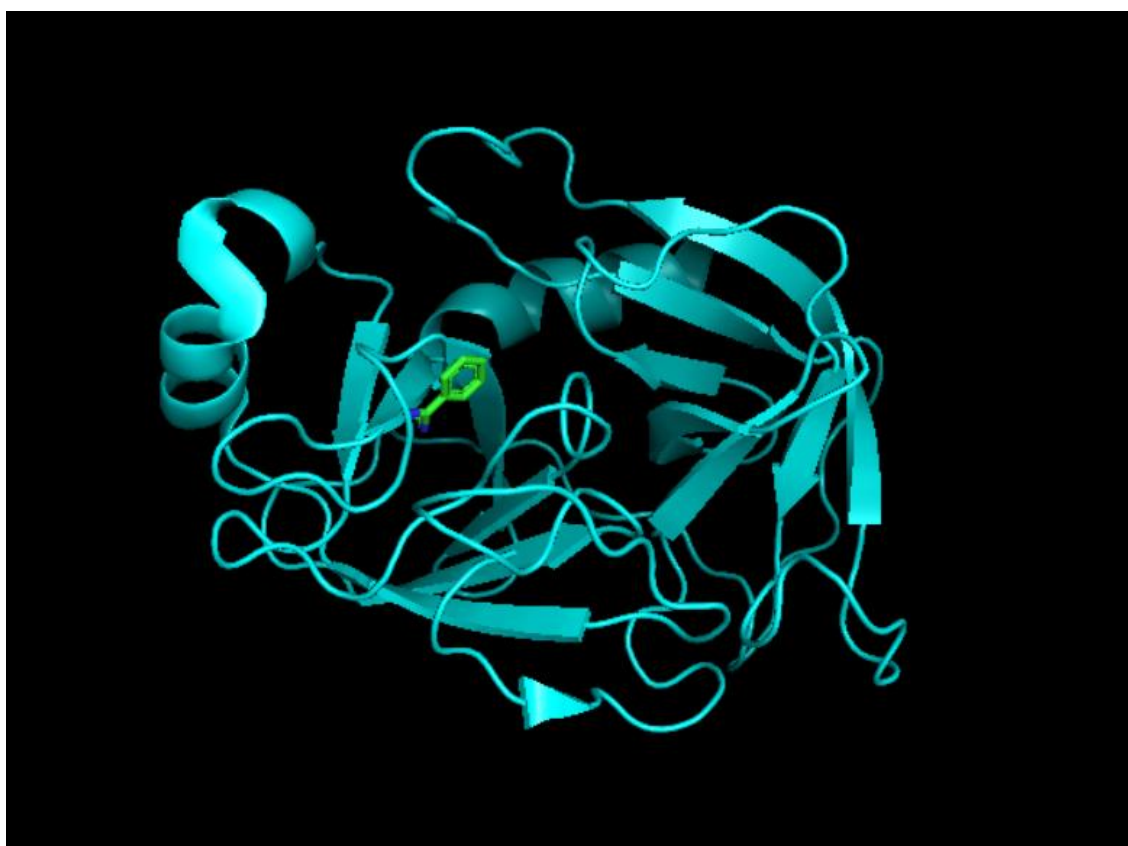


Figure 7. β -trypsin (PDB ID: 3PTB) structure. Cartoon representation created in *PyMOL* of β -trypsin shown as a cyan ribbon, with the co-crystallised benzamidine ligand displayed as green sticks within the S1 specificity pocket.

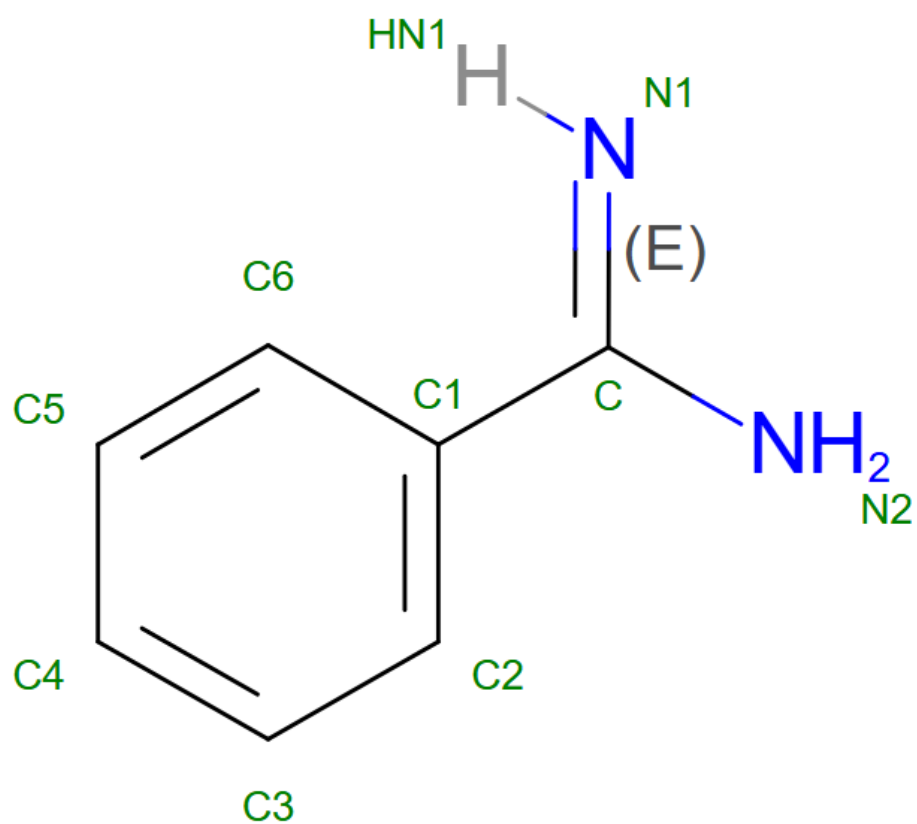


Figure 8. Two-dimensional chemical structure of benzamidine (3PTB ligand). The positively charged amidine group forms the critical electrostatic interaction with Asp189 in the S1 pocket.

3.4 Preparation

3.4.1 *AutoDock Vina* and *Smina*

Crystal structures were downloaded from the PDB and separated into protein and ligand components in *PyMOL*. Receptor preparation involved stripping crystallographic waters and non-essential chains, adding polar hydrogens, assigning Gasteiger charges, and exporting in PDBQT format. All receptors were treated as rigid throughout. Ligands followed the same charge assignment workflow; rigid and flexible files were generated separately by fixing or defining torsional bonds as appropriate. Grid boxes were centred on the crystallographic ligand position. Grid size (18–26 Å), exhaustiveness, number of output poses, and random seed were varied one at a time across experiments to probe parameter sensitivity while keeping conditions otherwise controlled.

3.4.2 *GOLD*

GOLD docking was carried out in Hermes (CCDC v2025.2), with the 3PTB complex loaded directly. Crystal waters within 5 Å of the ligand were retained; hydrogens were added automatically and charges assigned using *GOLD*'s default settings before export in MOL2 format. Ligand geometries were energy-minimised with the MMFF94 force field prior to docking. Binding site definitions were centred on the crystallographic ligand and kept spatially consistent with the *Vina/Smina* grid coordinates used elsewhere. All docking runs used the ChemPLP scoring function with 10 genetic algorithm runs per ligand.

3.4.3 Experimental Design Rationale

Parameter intervals were chosen to test meaningful variation in sampling depth and spatial restriction without straying into settings that would be unrealistic in practice. Exhaustiveness values of 8, 16, and 32 correspond roughly to low, moderate, and higher search effort within *Vina*-based engines. Grid dimensions of 18–26 Å were selected to span tightly ligand-centred boxes through to more permissive search volumes, giving a reasonable window onto spatial sensitivity. Three random seeds (12345, 54321, 99999) were included to capture stochastic variability across runs. In all parameter sensitivity experiments, only one variable was changed at a time, everything else held fixed to isolate each parameter's effect on predicted binding energy and avoid the interpretive ambiguity that comes with simultaneous multi-variable changes.

3.5 Docking Experiments

Docking was conducted systematically across three systems, 1HVR, 4PH9, and 3PTB, using consistent baseline grid placement. All runs were executed via the command line, and the top-ranked pose, known as mode 1, was selected for analysis. Parameter sensitivity analysis concentrated on binding energy only, while RMSD was used exclusively for pose reproduction and ligand modification assessment.

Experiment	Systems	Engines	Purpose	Primary Metric
1	4PH9	<i>Vina</i>	Preliminary workflow validation	Binding energy + RMSD
2	1HVR, 3PTB	<i>Vina</i> , <i>Smina</i>	Rigid vs flexible redocking	Binding energy + RMSD
3	1HVR, 3PTB	<i>Vina</i> , <i>Smina</i>	Parameter sensitivity (exhaustiveness, grid, seed)	Binding energy + RMSD
4	1HVR, 3PTB	<i>Vina</i> , <i>Smina</i>	Ligand modification tolerance	Binding energy + RMSD
5	3PTB	<i>Vina</i> , <i>Smina</i>	Interaction validation (ChimeraX)	Pharmacophore conservation
6	3PTB	<i>GOLD</i>	Commercial validation	ChemPLP + RMSD

3.5.1 Command-Line Docking Implementation

AutoDock Vina runs were automated using Python scripts executed within the native Windows environment. The subprocess module was used to call *AutoDock Vina* via the command line, enabling systematic parameter sweeps and controlled variation of exhaustiveness, grid size, and random seed. *Smina* runs were executed through Windows Subsystem for Linux (WSL) using a Linux-based command-line environment, guaranteeing compatibility with the compiled *Smina* binary and consistent parameter control equivalent to the *AutoDock Vina* workflow. Both environments produced PDBQT outputs that were analysed identically in subsequent RMSD and interaction assessments.

The complete command-line implementation, including representative docking outputs for both engines, is provided in Appendix A.

3.6 Evaluation Metrics

Docking performance was measured using three complementary metrics: binding energy score (kcal/mol) to assess comparative stability and parameter sensitivity within *Vina* and *Smina*; heavy-atom ligand RMSD relative to the crystallographic pose calculated in ADT; and hydrogen bond and contact profiling using ChimeraX to confirm pharmacophore conservation. Scores from different engines were interpreted within their native scoring scales and not compared directly in absolute magnitude. No formal statistical testing was performed; variability was assessed descriptively across controlled parameter differences.

3.6.1 Root Mean Square Deviation (RMSD)

To calculate RMSD relative to the experimental structure, the crystallographic complex and each top-ranked docked pose were loaded into *AutoDockTools* (ADT) and superimposed using heavy atoms only. All reported RMSD values, therefore, correspond to heavy-atom ligand RMSD relative to the crystallographic pose. RMSD values below 2.0 Å were considered successful reproduction of the experimental binding mode, with values near or below 1.0–1.5 Å interpreted as excellent agreement (Pagadala, Syed and Tuszynski, 2017). Higher RMSD values were taken to indicate incorrect or displaced poses, even when docking scores appeared favourable.

3.6.2 Pose Quality

Docked pose quality was assessed by comparing hydrogen-bond and contact interactions between the crystal and docked ligands using ChimeraX (Goddard *et al.*, 2018). FindHBonds identifies donor–acceptor pairs meeting geometric criteria (distance ≤ 3.5 Å, angle $\geq 120^\circ$).

FindClashes/Contacts detects non-bonded interactions, with clashes defined as less than 0.6 Å van der Waals overlap and contacts as ≤ 5 Å. Matching hydrogen-bond counts and identical key residue contacts indicate preserved pharmacophore geometry, whereas discrepancies reveal altered binding modes even when RMSD values remain acceptable.

3.6.3 Binding Energy Score

Binding energy scores (kcal/mol) were recorded for the top-ranked pose across all docking conditions. More negative values indicate stronger predicted binding. Scores were interpreted as relative ranking indicators within each engine rather than absolute thermodynamic values and were not compared directly among different engines or protein targets.

3.6.4 CHEMPLP Score

GOLD pose evaluation used the ChemPLP scoring function, which returns unitless fitness values not directly comparable to kcal/mol energies from *Vina* or *Smina*. The highest-scoring pose per condition was selected for analysis.

3.6.5 Interaction Analysis

Interaction analysis was performed exclusively for 3PTB, where the well-defined Asp189 and Gly219 pharmacophore provided a clear reference for assessing conservation. Hydrogen-bond and contact profiling was conducted using ChimeraX, comparing crystal and docked ligand interaction profiles to confirm preservation of key pharmacophore contacts beyond geometric RMSD agreement (Goddard et al., 2018).

4. DOCKING RESULTS

performance was analysed through four complementary analyses: rigid redocking validation, flexible docking parameter sensitivity experiments, ligand modification tests, and interaction profiling. These experiments were designed to assess pose accuracy, scoring reproducibility, and protocol sensitivity across three benchmark protein–ligand systems.

4.1 4PH9

4.1.1 *Vina* Performance Flexible vs Rigid

Rigid docking with *Vina* reproduced the crystallographic binding pose for 4PH9 with good accuracy, yielding an RMSD of 0.500 Å against the crystal structure. Predicted binding affinity was more favourable under rigid conditions (−9.3 kcal/mol) than with flexible docking (−7.7 kcal/mol). Flexible docking produced an RMSD of 1.146 Å, which falls within the accepted 2 Å threshold for successful pose reproduction, though geometric accuracy was noticeably reduced relative to the rigid result (Table 1).

Condition	Flexibility	Binding energy (kcal/mol)	RMSD (Å)
Flexible docking	flexible	−7.7	1.146
Rigid docking	rigid	−9.3	0.500
Crystal	N/A	0.0	0.000

Table 1. 4PH9 *Vina* baseline flexible vs rigid ligand docking. Rigid docking achieved good pose recovery (RMSD 0.500 Å); flexible docking remained within the accepted 2 Å threshold (RMSD 1.146 Å).

Parameter Sensitivity

Vina flexible docking for 4PH9 was largely stable across parameter variations. Binding energies held at −7.7 kcal/mol across all exhaustiveness and random seed conditions, with the only deviation arising at the smallest grid size (18 Å), which produced a slightly weaker score of −6.8

kcal/mol. This likely reflects the tighter spatial constraint reducing the available search volume rather than a genuine change in affinity (Table 2).

Parameter type	Setting	Binding energy (kcal/mol)
Exhaustiveness	8	-7.7
Exhaustiveness	16	-7.7
Exhaustiveness	32	-7.7
Grid size	18	-6.8
Grid size	22	-7.7
Grid size	26	-7.6
Random seed	12345	-7.7
Random seed	54321	-7.7
Random seed	99999	-7.7

Table 2. 4PH9 Vina flexible ligand parameter sensitivity scan. Binding energies were highly consistent across conditions; only the 18 Å grid size produced a weaker predicted affinity.

Metric	Value
Number of runs (n)	9
Mean (kcal/mol)	-7.51
Standard deviation (kcal/mol)	0.32

Table 3. 4PH9 Vina Flexible Parameter Sensitivity Summary

4.1.2 Smina Performance

Flexible vs Rigid

Smina docking for 4PH9 gave an unusual result. Flexible docking performed better than rigid docking in terms of both binding energy (-6.1 versus -5.8 kcal/mol) and pose accuracy (RMSD 1.733 versus 2.878 Å), with the rigid protocol failing to reproduce the crystal pose within the 2 Å threshold. This was an unexpected finding (Table 4).

Condition	Flexibility	Binding energy (kcal/mol)	RMSD (Å)
Flexible docking	flexible	-6.1	1.733
Rigid docking	rigid	-5.8	2.878
Crystal	N/A	0	0.000

Table 4. 4PH9 *Smina* baseline flexible vs rigid ligand docking. Flexible docking produced a more favourable binding energy and lower RMSD than rigid docking, contrary to the pattern observed across all other systems.

Parameter Sensitivity

Smina flexible docking for 4PH9 was highly consistent across all parameter variations. Binding energies stayed at −6.1 kcal/mol across every condition tested, with only seed 99999 producing any deviation at all (−6.2 kcal/mol) (Table 5).

Parameter type	Setting	Binding energy (kcal/mol)
Exhaustiveness	8	-6.1
Exhaustiveness	16	-6.1
Exhaustiveness	32	-6.1
Grid size	18	-6.1
Grid size	22	-6.1
Grid size	26	-6.1
Random seed	12345	-6.1
Random seed	54321	-6.1
Random seed	99999	-6.2

Table 5. 4PH9 *Smina* flexible ligand parameter sensitivity scan. Binding energies remained at −6.1 kcal/mol across all conditions; only seed 99999 produced a marginal deviation (−6.2 kcal/mol).

Metric	Value
Number of runs (n)	9
Mean (kcal/mol)	-6.11
Standard deviation (kcal/mol)	0.03

Table 6. 4PH9 *Smina* Flexible Parameter Sensitivity Summary. CV = 0.5%, the lowest observed across all systems and engines in this study

4.2 1HVR

4.2.1 *Vina* performance Flexible vs Rigid

Rigid docking reproduced the crystallographic binding pose for 1HVR more accurately than flexible docking, with a lower RMSD (0.607 Å versus 1.782 Å) and a more favourable predicted binding energy (−20.5 versus −14.2 kcal/mol). Both protocols fell below the 2 Å threshold, confirming successful redocking in each case (Table 7).

Condition	Flexibility	Binding energy score (kcal/mol)	RMSD (Å)
Flexible docking	flexible	−14.2	1.782
Rigid docking	rigid	−20.5	0.607
Crystal	N/A	0.0	0.000

Table 7. 1HVR *Vina* baseline flexible vs rigid ligand docking. Rigid achieved lower RMSD (0.607 Å vs 1.782 Å) and more favourable binding energy (−20.5 vs −14.2 kcal/mol).

Parameter Sensitivity

Flexible docking for 1HVR was largely consistent across parameter variations, with binding energies sitting at −14.2 kcal/mol for eight of the nine conditions tested. The only exception was seed 54321, which produced a noticeably weaker score of −9.3 kcal/mol. Changes in exhaustiveness and grid size had no meaningful effect on predicted affinity, suggesting that stochastic seed selection was the main source of variability here rather than sampling depth or search volume (Table 8).

Parameter type	Setting	Binding energy (kcal/mol)
Exhaustiveness	8	-14.2
Exhaustiveness	16	-14.2
Exhaustiveness	32	-14.2
Grid size	18	-14.2
Grid size	22	-14.1
Grid size	26	-14.2
Random seed	12345	-14.2
Random seed	54321	-9.3
Random seed	99999	-14.2

Table 8. 1HVR *Vina* flexible ligand parameter sensitivity scan. Eight of nine runs clustered tightly at −14.2 kcal/mol; seed 54321 outlier (−9.3 kcal/mol).

	Metric	Value
	Number of runs (n)	9
	Mean (kcal/mol)	-13.64
	Standard deviation (kcal/mol)	1.63

Table 9. 1HVR – *Vina* Flexible Parameter Sensitivity Summary (Binding Energy Only)

4.2.2 Smina Performance Flexible vs Rigid

Rigid docking with Smina reproduced the crystallographic pose for 1HVR with notably high accuracy, achieving an RMSD of just 0.101 Å against the crystal structure. Flexible docking performed considerably worse on both metrics, with an RMSD of 2.244 Å and a weaker predicted binding energy (−9.3 versus −14.1 kcal/mol) and fell outside the 2 Å threshold for acceptable pose recovery (Table 10).

Condition	Flexibility	Binding energy score (kcal/mol)	RMSD (Å)
Flexible docking	flexible	-9.3	2.244
Rigid docking	rigid	-14.1	0.101
Crystal	N/A	0	0.000

Table 10. 1HVR *Smina* baseline flexible vs rigid ligand docking. Rigid outperformed flexible (RMSD 0.101 Å vs 2.244 Å).

Parameter Sensitivity

Smina flexible docking for 1HVR showed considerably more variability than the *Vina* results, with binding energies ranging from −8.2 to −14.2 kcal/mol across the conditions tested. The weakest score came at exhaustiveness 8 (−8.2 kcal/mol), while higher exhaustiveness values and larger grid sizes tended to produce more favourable predictions. Unlike *Vina*, where results were largely stable across parameter changes, Smina appears more sensitive to how search parameters are set under flexible docking conditions (Table 11).

Parameter type	Setting	Binding energy (kcal/mol)
Exhaustiveness	8	-8.2
Exhaustiveness	16	-14.2
Exhaustiveness	32	-9.3
Grid size	18	-9.3
Grid size	22	-9.3
Grid size	26	-14.2
Random seed	12345	-14.2
Random seed	54321	-9.3
Random seed	99999	-14.2

Table 11. 1HVR *Smina* flexible ligand parameter sensitivity scan. Range -8.2 to -14.2 kcal/mol across parameter combinations.

Metric	Value
Number of runs (n)	9
Mean (kcal/mol)	-11.36
Standard deviation (kcal/mol)	2.72

Table 12. 1HVR – *Smina* Flexible Parameter Sensitivity Summary (Binding Energy Only)

4.3 3PTB

4.3.1 Vina Performance

Flexible vs Rigid

Rigid docking with AutoDock Vina reproduced the crystallographic binding pose for 3PTB accurately, with an RMSD of 0.205 Å against the crystal structure. Flexible docking also fell within the 2 Å threshold (RMSD 1.045 Å), though with a weaker predicted binding energy (-14.5 versus -17.3 kcal/mol). Both protocols achieved successful redocking, with rigid docking performing better on both metrics (Table 13).

Condition	Flexibility	Binding energy score (kcal/mol)	RMSD (Å)
Flexible docking	flexible	-14.5	1.045
Rigid docking	rigid	-17.3	0.205
Crystal	N/A	0	0.000

Table 13. 3PTB *Vina* baseline flexible vs rigid ligand docking. Rigid superior (RMSD 0.205 Å vs 1.045 Å).

Parameter Sensitivity

Flexible docking for 3PTB showed considerable variation across parameter conditions. Binding energies dropped from -14.5 kcal/mol at exhaustiveness 8 to -8.3 and -8.2 kcal/mol at higher settings. Grid size had an even stronger effect, ranging from -11.8 kcal/mol at 18 Å down to -1.9 kcal/mol at 22 Å and -4.2 kcal/mol at 26 Å. The 22 Å result in particular is striking and likely reflects the ligand being displaced away from the primary binding pocket rather than a genuine affinity prediction. Random seed also introduced variability, with seed 54321 giving -7.8 kcal/mol compared to -14.4 and -14.5 kcal/mol for the other two (Table 14).

Parameter type	Setting	Binding energy (kcal/mol)
Exhaustiveness	8	-14.5
Exhaustiveness	16	-8.3
Exhaustiveness	32	-8.2
Grid size	18	-11.8
Grid size	22	-1.9
Grid size	26	-4.2
Random seed	12345	-14.4
Random seed	54321	-7.8
Random seed	99999	-14.5

Table 14. Parameter sensitivity shows strong dependence on exhaustiveness and grid size, with favourable scoring only at exhaustiveness 8.

Metric	Value
Number of runs (n)	9
Mean (kcal/mol)	-9.51
Standard deviation (kcal/mol)	4.63

Table 15. 3PTB Vina Flexible Parameter Sensitivity Summary (Binding Energy Only)

4.3.2 Smina Performance Flexible vs Rigid

Rigid docking with Smina reproduced the crystallographic pose for 3PTB accurately, achieving an RMSD of 0.132 Å against the crystal structure. Flexible docking also fell just within the 2 Å threshold (RMSD 1.988 Å), though with a weaker predicted binding energy (-16.2 versus -19.8 kcal/mol). Both protocols achieved successful redocking, with rigid docking performing better on both metrics (Table 16).

Condition	Flexibility	Binding energy score (kcal/mol)	RMSD (Å)
Flexible docking	flexible	-16.2	1.988
Rigid docking	rigid	-19.8	0.132
Crystal	N/A	0	0.000

Table 16. 3PTB *Smina* flexible vs rigid ligand docking. Rigid outperformed flexible (RMSD 0.132 Å vs 1.988 Å; -19.8 vs -16.2 kcal/mol).

Parameter Sensitivity

Smina flexible docking for 3PTB showed the greatest variability of any system tested, with binding energies ranging from -7.5 to -20.1 kcal/mol across all conditions. Exhaustiveness scores dropped from -12.8 kcal/mol at setting 8 to -9.6 kcal/mol at both higher settings. Grid size produced a wider spread, from -7.5 kcal/mol at 18 Å up to -15.6 kcal/mol at 26 Å. Random seed was less variable overall, with two seeds giving -15.6 kcal/mol, though seed 99999 produced the highest score in the entire scan at -20.1 kcal/mol. The range observed here is the widest in this study and suggests that *Smina* scoring for 3PTB is particularly sensitive to how docking parameters are configured (Table 17).

Parameter type	Setting	Binding energy (kcal/mol)
Exhaustiveness	8	-12.8
Exhaustiveness	16	-9.6
Exhaustiveness	32	-9.6
Grid size	18	-7.5
Grid size	22	-10.2
Grid size	26	-15.6
Random seed	12345	-15.6
Random seed	54321	-15.6
Random seed	99999	-20.1

Table 17. 3PTB *Smina* flexible ligand parameter sensitivity scan. Range -7.5 to -20.1 kcal/mol across parameter combinations.

Metric	Value
Number of runs (n)	9
Mean (kcal/mol)	-12.96
Standard deviation (kcal/mol)	4.06

Table 18. 3PTB *Smina* Flexible Parameter Sensitivity Summary (Binding Energy Only)

4.4 Ligand Modifications

To assess sensitivity to ligand preparation choices, four modifications were applied to the crystallographic ligands before redocking under otherwise identical conditions. For 3PTB, the benzamidine ligand was energy-minimised using the MMFF94 force field and independently reprotonated. For 1HVR, an additional proton was added to the XK2 ligand to assess charge sensitivity, and an aromatic ring was removed to evaluate the effect of major pharmacophore disruption. Each modified ligand was redocked using both Vina and Smina, and the results were compared with the unmodified baseline.

4.4.1 Ligand Modifications Performed

The structural consequences of each modification applied to the XK2 ligand are shown in Figure 9. Aromatic ring deletion produced the most pronounced structural change, removing the flanking phenyl groups and reducing the ligand to its core scaffold. The addition of an extra proton is visible as explicit hydrogens on the ligand structure. These structural changes directly influenced docking outcomes, as reported in Section 4.4.2. For 3PTB, energy minimisation and reprotonation produced no visually discernible structural changes to the benzamidine ligand and are therefore not shown. Their impact on docking outcomes is assessed through RMSD and binding energy in Section 4.4.2.

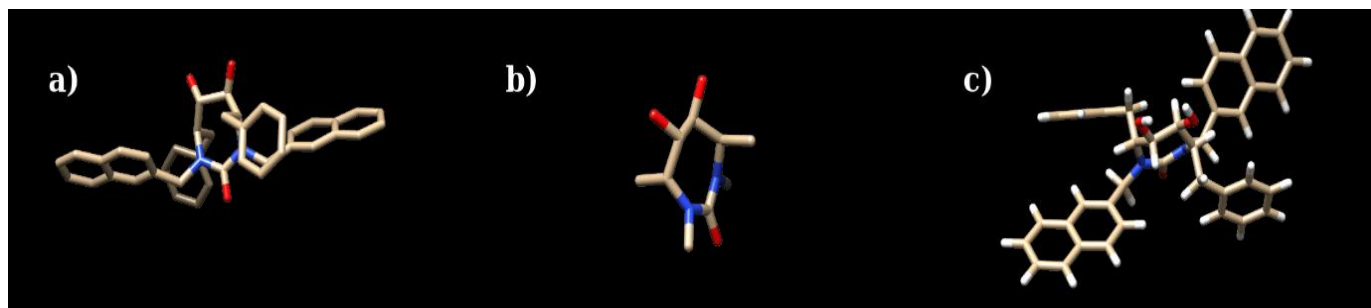


Figure 9. *PyMOL* representations of XK2 (1HVR) ligand modifications. (a) Unmodified XK2 crystal ligand prior to modification. (b) Aromatic ring deletion, which removes the flanking aromatic rings and reduces the pharmacophore to the core scaffold. (c) Extra proton addition is indicated by the presence of additional hydrogen atoms.

4.4.2 Results

Ligand modifications produced different effects depending on the system. For 3PTB, both docking engines tolerated the preparation changes well, maintaining near-native binding poses with RMSD values below 2 Å in all cases (Table 19). *Smina* produced slightly lower RMSD values than *Vina* (0.500–0.523 Å versus 0.735–1.204 Å), though predicted binding energies were similar for both engines at around –14 kcal/mol. For 1HVR, the picture was more mixed. *Vina* maintained accurate pose recovery after extra-hydrogen addition (RMSD 0.422 Å; –14.6 kcal/mol), but *Smina* failed to reproduce the crystallographic pose under the same conditions (RMSD 2.755 Å; –9.5 kcal/mol). Aromatic ring deletion caused pose failure for both engines, with RMSD values above 5 Å and weaker binding energies (–5.6 kcal/mol). This is expected given that removing the flanking phenyl groups eliminates the hydrophobic contacts that anchor the ligand within the binding pocket.

Complex	Modification	Engine	Binding Energy	RMSD (Å)
3PTB	Energy-minimised	<i>vina</i>	-14.1	1.204
3PTB	Energy-minimised	<i>smina</i>	-14.9	0.500
3PTB	Reprotonated	<i>vina</i>	-14.4	0.735
3PTB	Reprotonated	<i>smina</i>	-14.8	0.523
1HVR	Extra proton	<i>vina</i>	-14.6	0.422
1HVR	Extra proton	<i>smina</i>	-9.5	2.755
1HVR	Aromatic ring deletion	<i>vina</i>	-5.6	5.405
1HVR	Aromatic ring deletion	<i>smina</i>	-5.6	5.176

Table 19. Ligand modification experiments. 3PTB tolerates preparation changes (RMSD <1.5 Å); 1HVR ring deletion correctly fails (>5 Å RMSD).

4.4.3 Interaction Patterns

PyMOL interaction analysis confirmed preservation of the core hydrogen-bond network defining the 3PTB crystal pharmacophore across all ligand-modification conditions (Figure 10). Hydrogen-bond interactions with Asp189 and Gly219 were conserved in every case, indicating that the primary electrostatic interaction responsible for benzamidine recognition inside the S1 pocket remained intact regardless of ligand-preparation protocol. Hydrophobic contacts showed minor protocol-dependent variation but did not disrupt the overall interaction geometry. Together, these results support the geometric consistency of the docked poses despite modest differences in secondary contacts.

Complex	Condition	Key H-bonds	Hydrophobic Contacts	Chimera Match
3PTB	Crystal reference	Asp189 N1/N2, Gly219 N1	Ile138/162, Trp215/Gly216	Reference
3PTB	<i>Vina</i> Mod1	Asp189 OD1/OD2- N1/N2, Gly219 N1	None detected	Preserved
3PTB	<i>Smina</i> Mod1	Asp189 OD1-N1/N2, Gly219 N1	Ile138, Ala160, Ala183	Preserved
3PTB	<i>Vina</i> Mod2	Asp189 OD1/OD2- N1/N2, Gly219 N1	Ile138, Ile162, Ala160	Preserved
3PTB	<i>Smina</i> Mod2	Asp189 OD1-N1/N2, Gly219 N1	Ile138, Ile162, Ala160	Preserved

Table 20. 3PTB Chimaera interaction analysis. *Vina* and *Smina* preserve Asp189 N1/N2/Gly219 N1 pharmacophore across energy-minimised (Mod1) and reprotonated (Mod2) ligands.

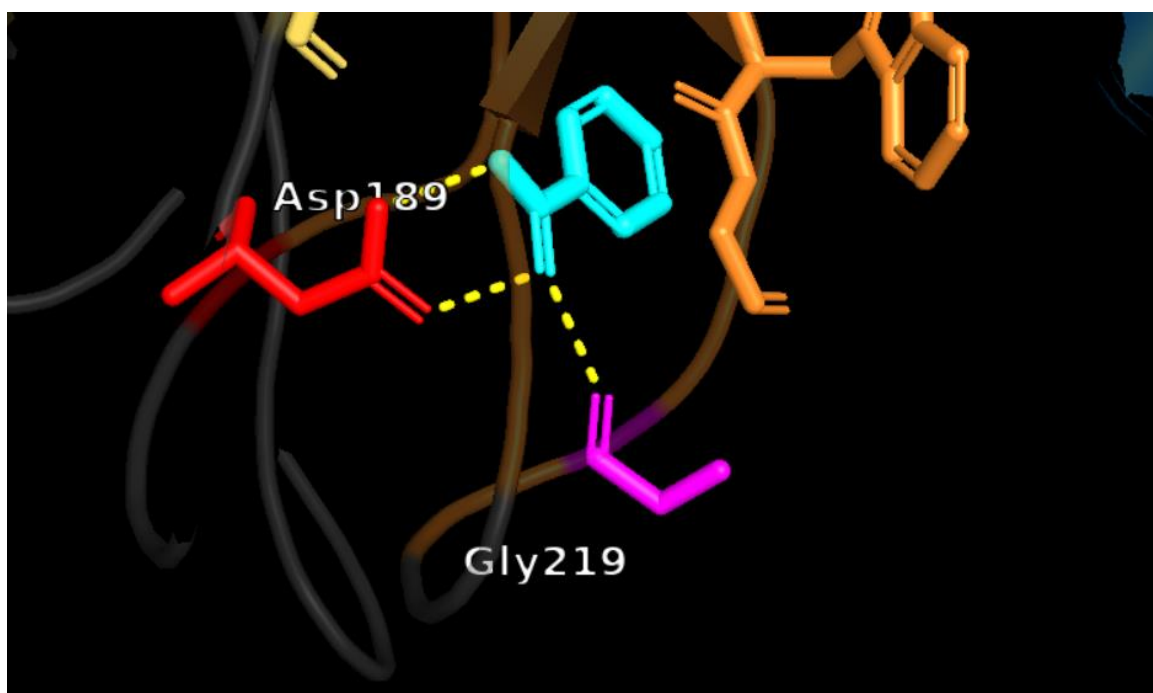


Figure 10. *PyMOL* visualisation of hydrogen bond interactions between benzamidine and the 3PTB S1 binding site. Yellow dashed lines indicate polar contacts between the ligand and key pharmacophore residues Asp189 and Gly219, confirming conservation of the primary interaction geometry in the docked pose.

4.4.4 Pose Reproduction Analysis

All three docking programs successfully reproduced the experimental 3PTB binding mode, with RMSD values below 2 Å relative to the crystallographic reference structure (Table 21). *Smina* produced the lowest RMSD (0.132 Å), followed by *Vina* (0.205 Å) and *GOLD* (0.642 Å), indicating accurate pose recovery across all engines. Because the scoring functions vary fundamentally between docking programs, with ChemPLP reporting unitless fitness values and the other methods using empirical kcal/mol estimates, score magnitudes were interpreted only within their own scoring frameworks. Consequently, the comparative evaluation focused on pose accuracy rather than the direct comparison of scoring values.

Program	Scoring Function	Top-ranked Score	Units	RMSD (Å)
<i>GOLD</i>	CHEMPLP	55.01	Fitness (unitless)	0.642
<i>Vina</i>	empirical scoring function	-17.3	kcal/mol	0.205
<i>Smina</i>	modified Vina scoring function	-19.8	kcal/mol	0.132

Table 21. Comparative docking performance of *Vina*, *Smina* and *GOLD* for 3PTB, showing top-ranked docking scores (native engine units) and RMSD of the top pose relative to the crystallographic ligand.

The crystal ligand (reference) is shown in grey. The docked poses are shown as follows: *GOLD* (CHEMPLP) in yellow, *AutoDock Vina* in pink/magenta, and *Smina* in cyan/blue. RMSD values were calculated using AutoDockTools (ADT), and PyMOL was used for structural visualisation after superposition.

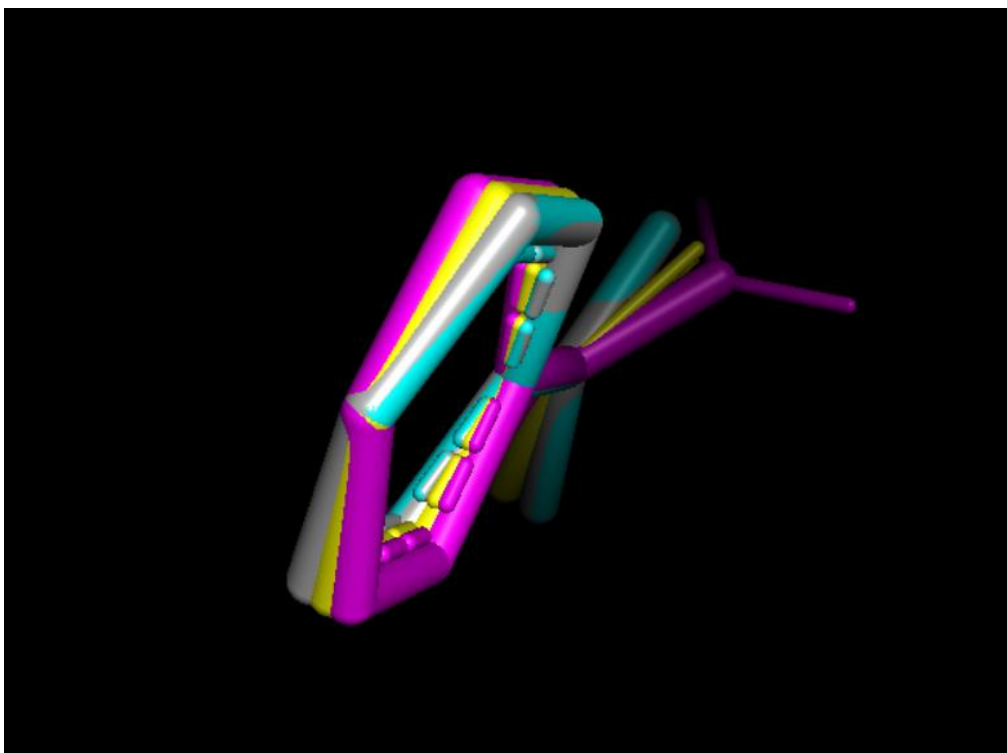


Figure 11. Overlay of docked poses with the crystal ligand.

Taken together, these results indicate that docking reproducibility depends on both the target and binding site and the search strategy employed. The following discussion considers these patterns in terms of binding-site architecture, parameter sensitivity, ligand preparation, and engine-specific scoring behaviour.

5. DISCUSSION

5.1 Rigid Versus Flexible Ligand Docking

Rigid redocking consistently outperformed flexible docking within pose accuracy among all three systems, as expected when the crystallographic conformation is already known: constraining torsional freedom reduces the search space and limits sampling of non-native orientations (Huang, 2018). The one exception, 4PH9 with *Smina*, suggests that inflexible protocols are not universally superior and may be sensitive to scoring function behaviour in specific binding environments. This is consistent with previous work showing that engine-specific differences appear even when fixed protocols are generally reliable (Alogheli *et al.*, 2017). Rigid docking, therefore, serves as a useful convergence baseline, but not an unconditional one.

Flexible docking increases the number of conformations that must be sampled during stochastic search. Although this more closely reflects biological ligand behaviour, it can also increase the chance of sampling non-native orientations that deviate from the crystallographic pose, a limitation well documented in the docking literature (Trott & Olson, 2010; Koes, Baumgartner and Camacho, 2013). In the redocking experiments, this additional flexibility introduced variability and did not consistently improve pose accuracy. One exception was observed for 4PH9 with *Smina*, where flexible docking produced a more favourable binding energy (-6.1 kcal/mol) and lower RMSD (1.733 Å) than rigid docking (-5.8 kcal/mol, RMSD 2.878 Å). This suggests that rigid *Smina* docking was less successful for this system under the conditions tested and may reflect the sensitivity of the scoring function and grid configuration applied to the COX-2 binding site. Consistent with previous benchmarking work showing that rigid protocols often perform well when the correct binding mode is well defined (Alogheli *et al.*, 2017; Huang, 2018). While also showing that engine-specific differences can occur.

5.2 Parameter Sensitivity and Target-Dependent Reproducibility

5.2.1 Structural Basis for Target-Dependent Sensitivity

The reproducibility differences between systems reflect the structural character of each binding environment rather than pocket size alone. In 3PTB, binding depends heavily on the electrostatic interaction between benzamidine and Asp189, so there is very little room for the ligand to be placed incorrectly. Unlike systems with more distributed interaction networks, small deviations here can lead to a large loss in score. This offers a reasonable explanation for the extreme parameter sensitivity observed and suggests that single-residue dominated binding sites may be a more challenging class of docking target than those with more distributed interactions.

The 1HVR system represents the opposite case. The near symmetry of the XK2 cyclic urea scaffold means that more than one ligand orientation can produce a similar docking outcome within the same energy terrain. As a result, the stochastic search may converge more reliably over various seeds and parameter settings. This could help explain the low variability observed for 1HVR across engines and configurations. However, this stability may reflect score degeneracy rather than true pose convergence. Similar binding energies can arise from more than one orientation, so consistent scores do not necessarily mean that the poses are identical. The intermediate behaviour of 4PH9 is less straightforward to explain. COX-2 exists as a homodimer, and docking was performed on chain A only. The absence of the homodimeric partner may have produced a partially constrained mixed interaction surface, although this remains unresolved in the present study. The very low variability of 4PH9 *Smina* (CV = 0.5%) compared with 4PH9 *Vina* (CV = 4.3%) may also reflect how the Vina scoring function responds to the hydrophobic character of the COX-2 active site. Overall, these results suggest that docking reproducibility depends not just on pocket size or constraint, but on the structural basis of the main binding interaction.

In practical terms, this means that a researcher who validates a docking protocol on one target and then applies the same settings to a structurally different system should not assume that the workflow will transfer cleanly. Settings that work well for one system may not produce the correct sampling geometry in another, and single docking jobs with default parameters should not be treated as definitive results without further validation. This has a direct practical consequence for how docking results are communicated. If a researcher reports a single docking score from a default run without disclosing the target's system sensitivity profile, that result is difficult to interpret or reproduce. This study suggests that reporting standards for docking should include, at a minimum, a seed variation test and a grid size sensitivity check, particularly for targets with electrostatically constrained binding sites (Bursulaya *et al.*, 2003). These are low-cost additions to any workflow that would substantially improve the reliability of reported results.

5.2.2 Anomalies

The fact that a lower exhaustiveness value of 8 produced the most favourable binding energy for 3PTB, while higher values of 16 and 32 led to less favourable scores, seems surprising at first. When considering the spike-like energy landscape discussed earlier, it makes sense. At lower exhaustiveness, the stochastic search makes fewer attempts, so it is more likely to find the strong electrostatic interaction with Asp189 before exploring other orientations. As exhaustiveness increases, more nearby minima are sampled, and some of these score well even if they do not represent the correct binding geometry. The main problem is not the amount of computational effort, but rather the scoring function's limited ability to rank a larger pool of poses correctly.

This was observed for 3PTB but not for 1HVR, where changing exhaustiveness had minimal impact on the score. That contrast supports the idea that scoring function mis-ranking under expanded search is more of a problem within systems with a single dominant minimum, where incorrect poses compete with one clearly correct solution. Increasing sampling exhaustiveness is only useful if the scoring function can distinguish the correct pose from near-native alternatives. When it cannot, increasing search depth may actually hinder convergence caused by introducing more incorrect poses. Rather than undermining the study, this finding supports the idea that 3PTB is a challenging docking target and suggests that default exhaustiveness settings should be validated rather than assumed.

A second unusual result was seen with 4PH9 Smina, where flexible docking outperformed rigid docking on both pose accuracy and predicted binding energy, which differed from every other result in the study. It is not entirely clear why this happened from the data alone. It could relate to ibuprofen having rotatable bonds, or to both protocols producing relatively weak scores for this

system, regardless of the flexibility setting. It is also possible that it was simply a random result from a single run, and more replicates would be needed to say anything more definitive.

Finally, random seed 54321 produced notably weaker scores in two separate systems, giving -9.3 kcal/mol in the 1HVR Vina scan against a cluster of -14.2 kcal/mol, and -7.8 kcal/mol in the 3PTB Vina scan against -14.4 and -14.5 kcal/mol for the other two seeds. The same seed underperforming across two different systems is worth noting, though with only three seeds tested per system, it is difficult to say whether this reflects a real pattern or coincidence.

5.2.3 Grid Size Effects

The particularly unfavourable binding energy at grid size $22 \text{ \AA}$ for 3PTB (-1.9 kcal/mol) was the worst result in the parameter sensitivity dataset and suggests a poor docking outcome for this condition. Benzamidine binding in the S1 pocket depends heavily on the precise positioning of the amidinium group relative to Asp189, so a grid that does not centre sampling properly on that interaction can reduce performance. The fact that $18 \text{ \AA}$ performed better than $22 \text{ \AA}$ is interesting because the tighter grid forces the search into the S1 pocket, while the intermediate $22 \text{ \AA}$ grid may have allowed sampling to drift away from Asp189 without being sufficiently restrictive to recover the correct pose. For this kind of target, grid definition appears to be more than just a technical detail.

5.2.4 Coefficient of Variation Analysis

Throughout this analysis, binding energy variance is used as an indicator of scoring consistency amid parameter unrest rather than as a measure of thermodynamic accuracy. The scores, therefore, reflect relative ranking stability within each system and should not be interpreted as physical binding affinities. 4PH9 *Smina* showed the lowest relative variability (CV = 0.5%), followed by 4PH9 *Vina* (CV = 4.3%), 1HVR *Vina* (CV = 11.9%), 1HVR *Smina* (CV = 23.9%), 3PTB *Smina* (CV = 31.3%), and 3PTB *Vina* (CV = 48.7%). This pattern suggests three different landscape regimes: a single-minimum constraint in 3PTB, a degenerate multi-minima landscape in 1HVR, and a partially constrained mixed-interaction surface in 4PH9. Overall, these results suggest that parameter sensitivity depends more on binding-site structure than on the docking engine alone. That supports the hypothesis that docking reproducibility would vary between protein targets and that the nature of that variation is structurally determined rather than purely stochastic. It also suggests that scoring function choice interacts with binding-site character in ways that are not always predictable, so engine selection should be treated as part of the experimental design rather than just a default choice.

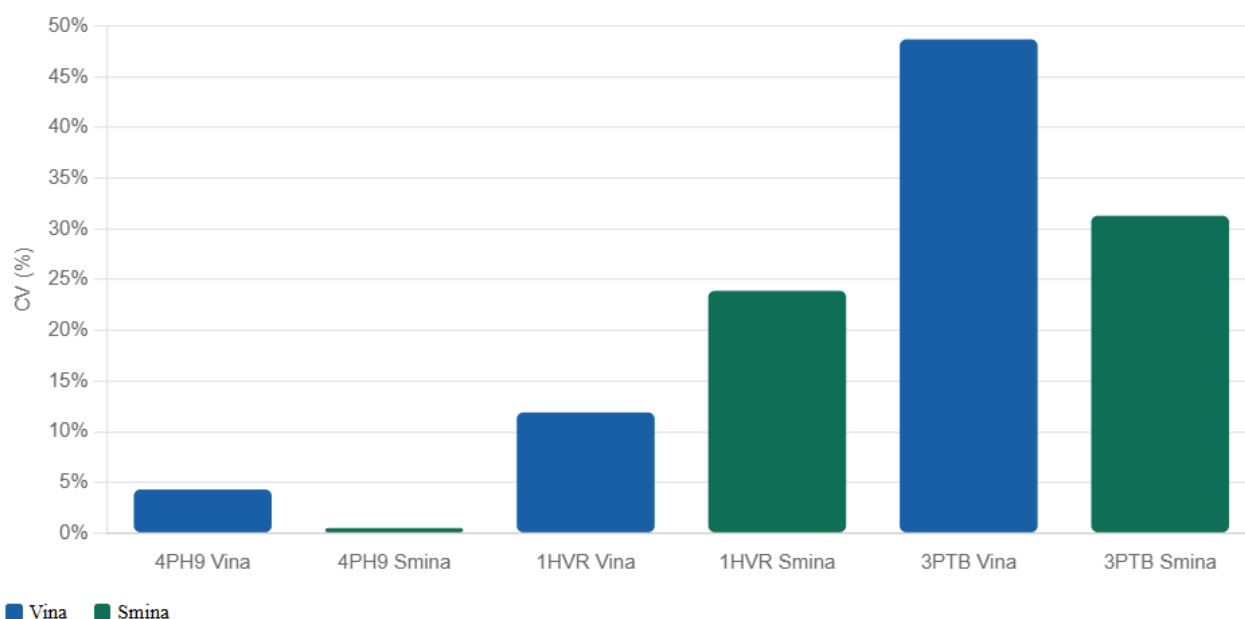


Figure 12. Coefficient of variation (CV%) for flexible docking binding energies across all benchmark systems and engines ($n = 9$ per condition). Lower CV = more reproducible.

5.3 Sampling Depth and Stochastic Effects

Variation under different random seeds shows that stochastic initialisation can affect docking search trajectories. Both *Vina* and *Smina* use stochastic global search followed by local optimisation steps, so when sampling depth is limited, independent runs may converge to different local solutions. For 3PTB, increasing exhaustiveness improved stability in some cases but did not eliminate variability, suggesting that sampling depth interacts with binding-site structure rather than acting alone. In practice, docking reproducibility should be assessed across multiple stochastic runs rather than a single execution. In this study, the 54321 seed outlier for 1HVR *Vina* (-9.3 kcal/mol versus -14.2 kcal/mol in the other conditions) was only detected because seed variation was tested directly. In a standard single-run workflow, that result would likely have been missed.

5.4 Ligand Preparation Sensitivity

Ligand modification experiments showed some engine-dependent behaviour. For 3PTB, energy minimisation and reprotonation were maintained near-native poses across both engines, suggesting that moderate preparation changes did not strongly disrupt the key interactions in the S1 pocket. In contrast, additional protonation in 1HVR caused the engines to diverge, with *Smina* failing to reproduce the crystallographic pose while *Vina* kept a near-native geometry. This is consistent evidence that protonation state and charge distribution can affect scoring terms and hydrogen-bond geometry, meaning even small preparation changes may alter docking outcomes (ten Brink and Exner, 2009). Although XK2 has multiple rotatable bonds and appears structurally flexible, reproducibility in 1HVR stayed relatively high across conditions because the urea anchor

seems to dominate the binding mode, limiting the effect of ligand flexibility on sampling. *Smina* also appears to be more sensitive than *Vina* in this setting, which may reflect its scoring behaviour and vulnerability to charge redistribution (Koes, Baumgartner and Camacho, 2013). Overall, the results suggest that extra flexibility in the software does not always improve performance and may, in some cases, make the workflow less robust.

5.5 Interaction Pattern Validation

RMSD alone does not guarantee preservation of key binding interactions. Interaction analysis using ChimeraX confirmed conservation of the Asp189 and Gly219 hydrogen bonds within the 3PTB system under valid docking conditions. Preservation of the interactions supports structural accuracy beyond simple geometric overlap (Goddard *et al.*, 2018).

Because docking scores are empirical approximations rather than rigorous free energy (Kitchen *et al.*, 2004). Combining RMSD, predicted binding energy, and interaction analysis appears to provide a stronger framework for validating docking results (Bursulaya *et al.*, 2003; Huang, 2018). No single metric alone was sufficient to confirm pose Quality across all conditions assessed.

This observation has a practical implication for how docking validation is reported. Many published studies rely on RMSD alone as the criterion for successful pose reproduction, applying the 2Å threshold as a binary pass/fail cutoff (Warren *et al.*, 2006). However, RMSD measures geometric displacement of the entire ligand and can mask local errors in pharmacophore geometry; a pose may fall within 2 Å overall while still misplacing a key functional group relative to a critical residue. For 3PTB, the conservation of Asp189 and Gly219 contacts across all modification conditions confirmed that the docked poses were not only geometrically close to the crystal structure but also chemically meaningful (Bursulaya *et al.*, 2003). In a prospective screening context, where the crystal pose is unknown, this kind of interaction profiling would be essential rather than supplementary. The results, therefore, support the use of a multi-metric validation framework as standard practice rather than an optional addition.

5.6 Comparison with Commercial *GOLD*

GOLD achieved pose reproduction accuracy comparable to that obtained using *Vina* and *Smina* for the 3PTB system. However, *GOLD* uses a genetic algorithm search strategy that differs from the stochastic optimisation methods used by *Vina*-derived (Jones *et al.*, 1997; Trott & Olson, 2010). All three programs converged on similar binding poses under controlled redocking conditions. This is consistent with previous benchmarking work (Bursulaya *et al.*, 2003; Huang, 2018). The performance differences between docking engines are often modest in single-ligand redocking experiments. Overall, the present findings support the reliability of carefully parameterised open-source docking workflows for structurally well-defined binding sites.

From a practical standpoint, this result was unexpected. Given that *GOLD* is a commercial package requiring institutional licensing, slightly better performance might have been anticipated. The fact that *Vina* and *Smina* produced comparable or lower RMSD values under matched conditions indicates that, for well-defined redocking tasks, the open-source tools were not necessarily at a disadvantage. It should be noted, however, that *GOLD* was only tested on a single system under time constraints, and a more comprehensive cross-target evaluation may have shown differences not captured here. Nevertheless, for academic researchers working without access to commercial software, these findings offer some reassurance that carefully applied open-source docking can achieve comparable geometric accuracy.

The convergence of all three engines on similar binding poses also has implications for how engine choice is justified in academic workflows. In many published studies, the selection of a docking engine is driven by availability or convention rather than systematic evaluation (Cross *et al.*, 2009). The present results suggest that for well-defined, rigid binding sites such as the 3PTB S1 pocket, engine choice may matter less than parameter configuration and preparation protocol. This shifts the practical burden from software selection toward workflow validation, a conclusion consistent with the broader findings found here. It should also be noted that the ChemPLP scoring function used by *GOLD* differs essentially from the empirical functions used by *Vina* and *Smina*, noted in section 1.5 in Figure 1, yet all three converged on the same binding geometry. This suggests that for structurally constrained targets, the energy profile is sufficiently well defined that multiple scoring approaches locate the same solution, whereas for more flexible or degenerate systems, the choice of scoring function may have a large influence on outcome (Cheng *et al.*, 2009; Huang, 2018).

5.7 Limitations

Three protein–ligand systems were evaluated in this study, spanning structurally diverse binding site architectures. The selected ligands XK2, ibuprofen and benzamidine represent different levels of molecular complexity, allowing comparison across contrasting docking scenarios. *Smina* parameter sensitivity data for 4PH9 were collected as a later addition to the primary dataset, meaning that full cross-engine comparison across all three benchmark targets is now available, although the 4PH9 *Smina* results should be interpreted alongside the atypical rigid docking behaviour observed for this system.

Docking scores represent empirical approximations rather than rigorous free-energy calculations (Kitchen *et al.*, 2004). No formal statistical hypothesis testing was performed, and variability was assessed descriptively. Given the exploratory nature of the study and the small number of parameter combinations tested, descriptive statistics including mean, SD and CV were considered appropriate for characterising variability trends across conditions. Redocking experiments represent favourable validation conditions and do not assess enrichment performance across large ligand libraries (Bursulaya *et al.*, 2003). Coefficient of variation analysis provides a useful indicator of relative variability but has inherent limitations when applied to small sample sizes. Formal statistical testing, such as ANOVA or the Mann–Whitney U test, would require larger datasets across multiple parameter combinations and protein systems to draw statistically sound conclusions.

Finally, commercial validation using *GOLD* was limited to the 3PTB system. Extension of the same analysis to 1HVR and 4PH9 would enable a more complete cross-engine comparison. *vina* Additionally, COX-2 docking was performed on chain A of the homodimeric complex only, which may not fully represent the native binding environment and could have contributed to the intermediate reproducibility observed for this system.

The most immediate extension would be applying the same parameter sensitivity analysis to a more extensive target set, particularly shallow or allosteric binding sites, where reproducibility may be harder to achieve. A direct comparison among empirical and machine-learning scoring functions, such as GNINA, under matched conditions would also help determine whether improved scoring translates into improved reproducibility or whether the same target-dependent variability persists regardless of the scoring approach (McNutt *et al.*, 2021).

6. CONCLUSIONS

This study set out to evaluate whether docking reproducibility varies between protein targets under controlled parameter changes, and the results clearly demonstrate that it does. The three systems behaved differently in ways that reflect their binding-site architecture rather than the choice of software. 3PTB showed the greatest sensitivity overall, with binding energies varying substantially depending on exhaustiveness and grid size. This is likely attributable to the dependence of benzamidine binding on a single electrostatic contact with Asp189; when that interaction is missed, the score drops dramatically. 1HVR was considerably more stable across conditions, consistent with the more forgiving hydrophobic cavity and the symmetric XK2 scaffold, which can adopt multiple orientations with similar scores. 4PH9 occupied an intermediate position between these two extremes.

Rigid redocking performed better than flexible docking under nearly every condition, attesting to its value as a convergence baseline before moving to more complex protocols. The comparable RMSD values produced by GOLD, *Vina*, and *Smina* under matched conditions denote that carefully parameterised open-source workflows are not necessarily at a disadvantage for well-defined redocking tasks. The wider implication is that default parameter settings should not be applied without some form of target-specific validation. In a real drug discovery context, reliance upon a single default run could lead to a misranked compound being carried forward or a valid hit being discarded. This concern extends to machine-learning scoring functions, where training data derived from poorly validated docking workflows would propagate the same target-dependent inconsistencies into the model.

For future work, the most logical extension would be to apply the same parameter sensitivity analysis to a wider range of targets, particularly those with shallow or allosteric binding sites. Repeating the GOLD comparison on 1HVR and 4PH9 would also allow more complete conclusions about engine differences than the single-system comparison presented here. Allowing receptor flexibility through ensemble docking would be a natural further step, especially for HIV-1 protease, where flap movement is well documented and rigid receptor protocols cannot entirely capture the native binding environment.

Taken together, the findings of this study argue for a change in how reproducibility is treated in docking workflows from an assumption to a measurable quantity that should be evaluated as part of standard protocol development. The consistency of results across engines for well-defined targets and the pronounced variability for constrained single residue systems suggest that reproducibility is a property of the target as much as the software (Cheng *et al.*, 2009). Establishing this principle on a broader scale, across more diverse target classes and with larger parameter grids would provide the field with clearer guidance on when default settings can be trusted and when they cannot (Warren *et al.*, 2006; Cross *et al.*, 2009)

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APPENDIX

```
bash  
  
vina \  
--receptor receptor.pdbqt \  
--ligand ligand.pdbqt \  
--center_x CX \  
--center_y CY \  
--center_z CZ \  
--size_x SX \  
--size_y SY \  
--size_z SZ \  
--exhaustiveness EX \  
--seed SEED \  
--num_modes 9 \  
--out output.pdbqt \  
--log log.txt
```

Figure A1. Representative AutoDock Vina command structure used for parameter sweep experiments, showing the flags and placeholder variables for grid centre coordinates, grid size, exhaustiveness, and random seed across all three systems

